

How Do Vectors Add?

SYNOPSIS

Displacement and velocity are vector "counterparts" of distance and speed — they carry a direction as well as a size. Most quantities you'll meet are one or the other, vector or scalar, with no matching partner. Scalars add the way ordinary numbers do. Vectors don't: a vector is an arrow, and arrows add by *joining the head of one to the tail of the next* — the head-to-tail rule. A negative sign flips a vector's head and tail, pointing it the opposite way, which is how vector subtraction works. In this note we learn to add and subtract vectors diagrammatically; multiplication comes later.

PERFORMANCE EXPECTATIONS By the end of this note, you will be able to...

- PE1 Distinguish scalar addition from vector addition** — scalars add like ordinary numbers; vectors add head-to-tail.
- PE2 Apply the head-to-tail rule** to join two or more vectors, keeping each vector's own magnitude and orientation.
- PE3 Recognize that a negative sign flips a vector's head and tail** — it reverses direction without changing length. That's how subtraction works.
- PE4 Draw the resultant vector** — from the tail of the first vector to the head of the last — for two or more vectors joined head-to-tail.
- PE5 Determine a resultant's magnitude and direction by measurement** (ruler and protractor) for vectors joined at any angle.
- PE6 Recognize when vectors must be rejoined** — two vectors drawn joined at the tail (or head), or separated in space, must be moved head-to-tail before you can add them.



Relate the angle at the joint to the resultant's length — straight ahead gives the longest resultant, doubled back the shortest, and every angle between gives something in between.



You already know this

Head-to-tail addition isn't new — you've been doing it without the vocabulary:



Turn-by-turn directions

"3 blocks east, then 2 blocks north." Each leg starts where the last one ended — that's head-to-tail — and where you end up is the resultant.



Game-controller movement

Press right, then up. Your character's net move is the head-to-tail sum of the two presses, even though you made them one at a time.



Walking it back

Walk 5 blocks east, then 5 blocks west — the second leg is the *negative* of the first. You end up exactly where you started.

In this note, we work through

- I. What's the difference between a scalar quantity and a vector quantity?
- II. How do vectors add — and why can't we just add their lengths like numbers?
- III. What is the head-to-tail rule, and what does a negative sign do to a vector?
- IV. How do we find a resultant vector's magnitude and direction?
- V. How do we add vectors that are drawn joined at the tails — or separated in space?
- VI. How does the angle where two vectors join change the resultant's length?

VII. Which pairs of vectors produce identical resultants?



arrows join head to tail

Play with it: the head-to-tail lab

A and **B** start out **separated in space** — the way force vectors are usually drawn on a diagram. They can't be added as they sit there. Press **Join head-to-tail** to slide the second one across (watch it — it never rotates) until its tail meets the first one's head, then press **Show resultant**.

Three things to try. Switch to **A – B** and watch the negative sign flip B right around — a ghost of the old +B stays behind so you can see the flip. Press **Swap order** to draw B first instead: the resultant lands in exactly the same place, every time. And swing a direction slider while watching the **joint angle** chip — that angle is what decides how long the resultant comes out.

► INTERACTIVE · HEAD-TO-TAIL VECTOR ADDITION

Vector A — length

4.0 m

Vector A — angle

0°

Vector B — length

3.0 m

Vector B — angle

90°

A + B

A - B

→← Join head-to-tail

Show resultant

↔ Swap order

Show measurement

↺ Reset

|A| = 4.0

|B| = 3.0

joint angle = —

angle between A and B = —

Resultant length ≈ —

Resultant angle ≈ —

Vectors separated in space can't be added as drawn — slide one across (keeping its orientation!) until they connect head-to-tail.

Predict first: two vectors, 3 m and 4 m long, are joined head-to-tail at some angle. Which of these magnitudes is *impossible* for the resultant?

7 m

5 m

1 m

8 m

Watch: working through the note

WALKTHROUGH VIDEO · ~10 MIN

Video embed goes here — print the guided notes below before you press play.

The joint angle at a glance

When two vectors of lengths a and b connect head-to-tail, the angle at the joint is what sets the resultant's length. Two different angles get quoted for the same picture, so the table lists both — read the column that matches the question you're answering:

HOW THEY JOIN	JOINT ANGLE at the corner	ANGLE BETWEEN A and B	RESULTANT LENGTH
Straight ahead — same direction	180°	0°	$a + b$ — the longest possible
At a corner — perpendicular	90°	90°	the diagonal of the corner — between the two extremes
Doubled back — opposite directions	0°	180°	$a - b$ — the shortest possible
Doubled back, equal lengths	0°	180°	zero — you end where you started
The joint angle is the one you measure on the diagram, at the corner where the arrows meet — that's the convention used in our guided notes and in the lab above.			

HOW THEY JOIN

**JOINT
ANGLE**
at the
corner

**ANGLE
BETWEEN**
A and B

RESULTANT LENGTH

The **angle between A and B** is the one most textbooks and WebAssign quote; it's the supplement of the joint angle, so the two always add to 180° .

Looking ahead: here we add vectors by *drawing* — ruler, protractor, grid paper. In Note 2 we run the rule in reverse and *resolve* one vector into a horizontal and a vertical piece, and in Note 2a trigonometry turns those drawings into calculations. Master the picture first; the math is just the picture with numbers.



tail of the first → head of the last

Three rules to remember



Head-to-tail

Join the head of one vector to the tail of the next — rejoining them first if they're drawn tail-to-tail or apart, and never rotating them.



C: first tail → last head

The resultant starts at the tail of the first vector and ends at the head of the last — whichever order you draw them in.



−B flips B

A negative sign swaps a vector's head and tail; its length never changes. Subtraction is just addition of the flipped vector.



Exit ticket

Q1. A vector quantity is best described as one with...

- ☐ Magnitude only.
- ☐ Direction only.
- ☐ Both magnitude and direction.
- ☐ Neither — it's just a number.

Q2. To add two vectors, you must join them...

- ☐ Tail-to-tail.
- ☐ Head-to-head.
- ☐ Head-to-tail — the resultant runs from the tail of the first to the head of the last.
- ☐ Side by side, without moving either one.

Q3. Two vectors, 5 m and 3 m, point in exactly opposite directions (180° apart) and are joined head-to-tail. What is the magnitude of the resultant?

m

Q4. Two force vectors are drawn joined at their tails. Before you can add them, you must...

- ☐ Add them exactly as drawn.
- ☐ Flip one of them.
- ☐ Separate them and rejoin them head-to-tail, keeping each vector's orientation.

☐ Measure the angle between them first.

Check my answers

Clear

Your notes

BEFORE THE VIDEO

Guided notes — blank (PDF)

Print or open on a tablet and work through the grid-paper vector exercises as you follow along.

AFTER THE MODULE

Completed notes (PDF)

Released when Module 2 closes — check your work against it.